

SERS Substrate-Agnostic Dataset Matrix

The corrected substrate-family analysis groups AgNP with Ag and AuNP with Au. Canonical chemical labels are used, with BT merged into benzenethiol. The table below shows the current number of spectra for each grouped chemical-substrate family pair.

Table 1: Current corrected grouped SERS matrix.

Chemical	Ag	Au	PICO	pSERS	Total	Note
4np	25	0	25	25	75	Au is N/A
benzenethiol	25	25	25	25	100	complete
pyridine	25	25	25	25	100	complete
DMF	0	228	0	0	228	Au only

The minimum recommended expansion is to add another responsive chemical on all four substrate families before heavily increasing spectra from already-covered cells.

Table 2: Minimum target matrix for stronger substrate-agnostic evaluation.

Chemical	Ag	Au	PICO	pSERS	Total	Priority
4np	25+	N/A	25+	25+	75+	already valid
benzenethiol	25+	25+	25+	25+	100+	already complete
pyridine	25+	25+	25+	25+	100+	already complete
additional chemical	25+	25+	25+	25+	100+	highest value

If time allows, the stronger target is two to three independent preparations or maps per chemical-substrate-family pair, with approximately 25 spectra per preparation. Independent preparations are more useful than many correlated spectra from a single map.